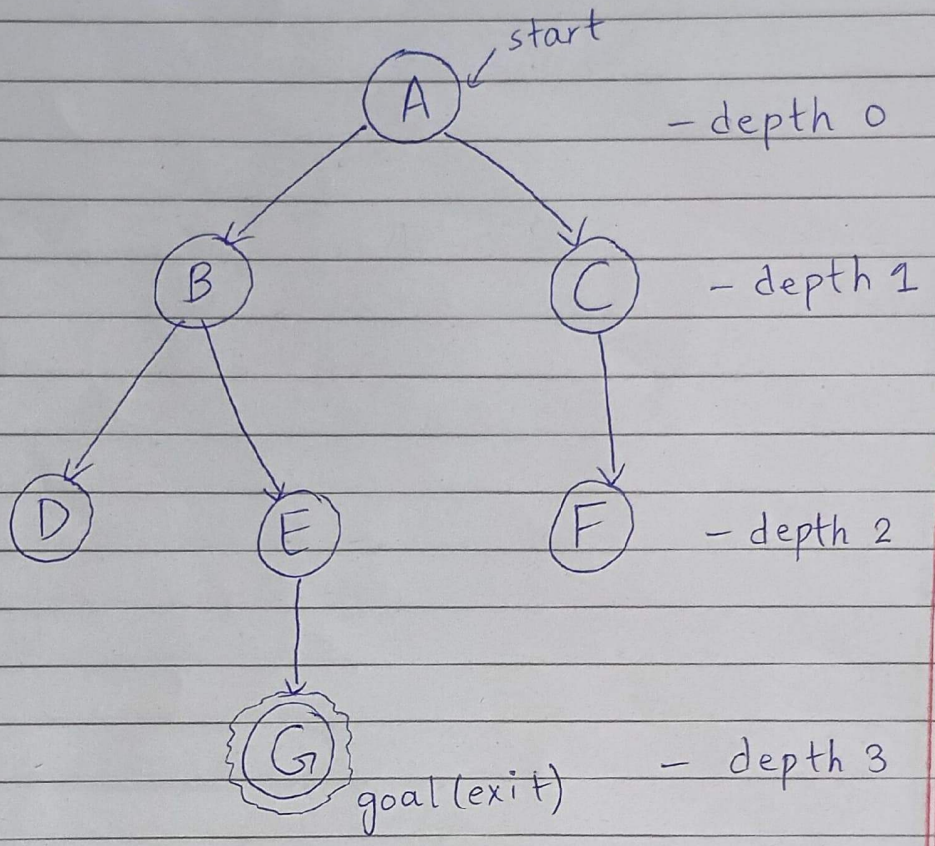
Date: Day:

Lab 3 - Task: 1 BFS & DFS Trace

Start = A Goal = G (graph from Section 3.2)

The graph:



successor order (left to right) = tie-breaker:

A: [B, C] B: [D, E] C: [F] D: [] E: [G]

F: [] G: []

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1) BFS Trace (queue, FIFO)

queue written front → back. Remove from the front, add children at the back.

Step	Processed	Path	Queue after step
start	–	–	A
1	A	A	B, C
2	B	A → B	C, D, E
3	C	A → C	D, E, F
4	D	A → B → D	E, F
5	E	A → B → E	F, G
6	F	A → C → F	G
7	G (goal!)	A → B → E → G	stop

Visit order: A, B, C, D, E, F, G
Final path: A → B → E → G
Nodes processed = 7

Notes:

- BFS goes level by level (depth 0, then 1, then 2...)
- It wasted time on C and F, they are not on the answer path.
- Queue held up to 3 nodes at once → BFS needs more memory when the graph is wide.

Date: Day:

2) DFS Trace (stack, LIFO)

stack written bottom → top. The RIGHTMOST item is popped next.

step	Processed	Path	stack after step
start	-	-	A
1	A	A	C, B
2	B	A → B	C, E, D
3	D	A → B → D	C, E
4	E	A → B → E	C, G
5	G (goal)	A → B → E → G	C (unused)

Visit order: A, B, D, E, G
Final path : A → B → E → G
Nodes processed = 5

Why reversed ()?

A stack gives back the LAST item pushed first. A has children [B, C]. If I push B then C, C is on top
→ C would be processed first (wrong order). So I push in reverse: C first, then B. Now B is on top
→ B is popped first = first listed neighbor first.
Example at step 1: push C, push B → stack = [C, B] (top = B)

Date:_____ Day:_____

3) Comparison & Code Check

Algo	Final path	Visit order	Count
BFS	A → B → E → G	A B C D E F G	7
DFS	A → B → E → G	A B D E G	5

Same path, but different search behaviour!

→ a path alone does not show HOW the algorithm searched.

Verified with the supplied code/output I actually got:

bfs - order ('A', 'G') → path ['A', 'B', 'E', 'G']
order ['A', 'B', 'C', 'D', 'E', 'F', 'G']

dfs - order ('A', 'G') → path ['A', 'B', 'E', 'G']

My hand trace matches the program output. Yes

Trigger Q: Why does BFS need more memory?

BFS keeps a whole level in the queue at once.

DFS keeps only the current path + waiting siblings.

Observation (from my lab 4.py photo):

My lab 4 DFS has no reversed (), so it visited C before B.

Successor order changes the DFS visit order.